UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT 101: ALGEBRA AND TRIGONOMETRY

TUTORIAL SET 3.

(1.) A polynomial $Q(x) = 2x^4 + 15x^3 + 27x^2 - 27x - 81$. Verify that $x = -3$ is a zero of $Q(x)$ and find its multiplicity.

(2.) Divide $P(x) = x^4 + 9x^2 + 2x - 1$ by $D(x) = x^2 + 4x + 1$.

(3.) For each of the following functions, determine the domain and range. Examine the behaviour of the function, and hence sketch the graph of the following in the same plane.

(a) $\frac{x^2-x-6}{x+1}$

(b) $\frac{2x}{x^2-4}$

(c) $\frac{2}{1-x^2}$